

PABLO ALBERTO SANTANA FLORES

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PROFESSIONAL SUMMARY

Chemical Engineer, Data Scientist, and PhD Candidate in Marine Sciences with 16+ years of expertise in quality assurance (QA/QC), analytical chemistry, and precision instrumentation. Currently excelling as a Chemistry & Lab Data Specialist at TELUS Digital, validating AI-generated molecular structures and refining safety protocols. Holds an M.Sc. in Sciences, alongside advanced certifications in Data Analytics, Artificial Intelligence, and HACCP compliance. A highly adaptable, detail-oriented professional with a proven track record in cross-functional collaboration, bridging the gap between raw chemical data and automated machine learning solutions.

PROFESSIONAL EXPERIENCE

TELUS Digital

Chemistry & Lab Data Specialist

Mérida, Mexico (Remote)

Feb 2026 – Present

- Validate AI-generated chemical formulas and molecular structures to ensure absolute accuracy and regulatory compliance.
- Audit and refine safety protocols for automated chemical recommendations and generative AI outputs.
- Structure and manage complex experimental datasets to optimize machine learning model training and predictive analytics.
- Translate advanced chemical concepts and data insights into actionable strategies for non-technical stakeholders.

A.S.T.M.

Quality Control Laboratory Analyst

Mérida, Mexico

Jan 2024 – Jan 2026

- Executed rigorous QA/QC protocols in strict adherence to ASTM and NMX industry standards.
- Spearheaded process optimization initiatives that successfully reduced operational non-conformities by 12%.
- Performed comprehensive physicochemical testing and maintained calibration of high-precision analytical instrumentation.

United Bakery Manufacturers

Quality Control & Process Technician

Ontario, Canada

Oct 2021 – Oct 2023

- Managed dynamic sampling programs and ensured strict compliance with Good Manufacturing Practices (GMP).
- Authored technical documentation and facilitated operational communication across cross-functional English-speaking teams.

CINVESTAV Unidad Mérida

R&D Master's Researcher

Mérida, Mexico

Jan 2018 – Sep 2021

- Led advanced material characterization utilizing FESEM, XRD, FTIR, and Raman spectroscopy.
- Collaborated on international research projects, delivering high-impact scientific insights and technical reports.

EDUCATION

PhD Candidate in Marine Sciences <i>CINVESTAV Unidad Mérida</i>	2018 – 2021
M.Sc. in Sciences <i>CINVESTAV Unidad Mérida</i>	2014
B.S. in Chemical Engineering <i>Universidad Autónoma de Coahuila (UAdeC)</i>	2009

CERTIFICATIONS

Google AI Professional Certificate (7-Course Sequence) <i>Google / Coursera</i>	2026
Analyze Data to Answer Questions (Data Analytics) <i>Google / Coursera</i>	2026
AI and Productivity <i>Santander Open Academy</i>	2025
HACCP System Certification <i>Secretaría de Salud del Estado de Nuevo León</i>	2025
Intro to Programming <i>Kaggle</i>	2023

TECHNICAL SKILLS & COMPETENCIES

- **Programming & Analytics:** Python, Data Analytics, SPSS, Mathematical Modeling, Machine Learning.
- **Engineering & Quality:** Analytical Instrumentation (Calibration & Validation), Quality Assurance (QA/QC), HACCP, GMP.
- **Core Competencies:** Cross-functional Team Leadership, Effective Communication, Adaptability, Proactive Problem Solving.
- **Languages:** English (C1 Full Professional Proficiency), Spanish (Native).